EUCAST European Committee on Antimicrobial Susceptibility Testing

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EUCAST recommendations for MIC determination of aerobic and facultative anaerobic bacterial species with broth microdilution

The EUCAST recommendations for MIC determination for non-fastidious organisms are in complete agreement with the recommendations from the International Standards Organisation (ISO). The broth microdilution method is described in **ISO 20776-1 (2019), Susceptibility testing of infectious agents and evaluation of performance of antimicrobial susceptibility test devices - Part 1: Broth micro-dilution reference method for testing the in vitro activity of antimicrobial agents against rapidly growing aerobic bacteria involved in infectious diseases.** This document is available for purchase from ISO or national standards organisations.

For fastidious organisms, EUCAST recommends the same methodology but with the use of MH-F broth (Cation-Adjusted Mueller-Hinton broth with 5% lysed horse blood and 20 mg/L beta-NAD), see Media preparation for EUCAST disk diffusion testing and for determination of MIC values by the broth microdilution method.

Most MIC endpoints are read at the lowest concentration of the agent that completely inhibits visible growth, as stated in ISO 20776-1. However some antimicrobial agents or organism-agent combinations need specific reading instructions, see EUCAST reading guide for broth microdilution.

This document describes the testing conditions for broth microdilution MIC determination, including methodological remarks, for bacterial species with EUCAST breakpoints, or epidemiological cut-off values (ECOFFs) in the EUCAST MIC database.

For MIC determination of *Mycobacteria*, see EUCAST reference method for MIC determination of anti-tuberculous agents.

Summary of EUCAST recommendations for broth microdilution of aerobic and facultative anaerobic bacterial species

Table 1a. Organisms tested in Cation-Adjusted Mueller-Hinton Broth (CAMHB)

Organism	Medium[1]	Inoculum[2]	Incubation[3]	Remarks[4]
Achromobacter spp.	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Acinetobacter spp.	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Aeromonas spp. (except *A. salmonicida sp. salmonicida*)	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Aeromonas salmonicida sp. salmonicida	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, **22±1ºC, 48±2h**	
Bacillus spp.	CAMHB	$5x10^5$ CFU/mL[5]	Sealed panels, air, 35±1ºC, 18±2h	For *B. anthracis*, see Note 5
Bordetella bronchiseptica	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Brucella melitensis	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, **48±2h**	
Burkholderia pseudomallei	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Enterobacterales	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Enterococcus spp.	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Pseudomonas spp.	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Staphylococcus spp.	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Stenotrophomonas maltophilia	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Vibrio spp.	CAMHB	$5x10^5$ CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	

[1] For information on supplementation or modification of the broth for specific antimicrobial agents, see **Table 2**.
[2] The final inoculum should give a final cell number of $5x10^5$ CFU/mL (target range $2x10^5$ to $8x10^5$ CFU/mL); viable counts should be performed to ensure that the final inoculum is within the specified range (see ISO 20776-1). For all microorganisms, the number of viable cells in the final inoculum depends on the growth phase of the initial culture. This effect is most pronounced for fastidious organisms, which normally needs at least two passages on suitable agar media when retrieved from freezer storage. The dilution required from a McFarland 0.5 suspension depends upon the bacterial species being tested and the method used for inoculum delivery. For *B. anthracis*, see Note 5.
[3] 24 h incubation for glycopeptides and for oxacillin when tested for *Staphylococcus* spp.
[4] For reading instructions, see EUCAST Reading Guide for broth microdilution. For antimicrobial agents with specific reading instructions, see **Table 3**.
[5] *Bacillus anthracis*: One CFU for this bacterium corresponds to a chain consisting of multiple cells rather than to a single cell. The inoculum should be based on dilutions made from the theoretical CFU/mL of a McF 0.5 solution (1-$2x10^8$CFU/mL) to reach a theoretical inoculum of $5x10^5$ CFU/mL.

Table 1b. Organisms tested in Cation-Adjusted Mueller-Hinton Broth + 5% lysed horse blood and 20 mg/L β-NAD (MH-F broth)

Organism	Medium[1]	Inoculum[2]	Incubation[3]	Remarks[4]
Actinobacillus pleuropneumoniae	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Aerococcus spp.	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	40-44 h incubation if insufficient growth after 16-20 h incubation.
Campylobacter coli and *C. jejuni*	MH-F broth	5x10^5 CFU/mL	Microaerobic environment, **41±1ºC, 24±1h**	Panels must be covered with a perforated seal. 40-48 h incubation if insufficient growth after 24 h incubation.
Corynebacterium spp.	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	40-44 h incubation if insufficient growth after 16-20 h incubation.
Haemophilus influenzae	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Kingella kingae	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	40-44 h incubation if insufficient growth after 16-20 h incubation.
Listeria monocytogenes	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Mannheimia haemolytica	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Moraxella catarrhalis	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Pasteurella spp.	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Streptococcus spp.	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	
Trueperella pyogenes	MH-F broth	5x10^5 CFU/mL	Sealed panels, air, 35±1ºC, 18±2h	

[1] For information on supplementation or modification of the broth for specific antimicrobial agents, see **Table 2**.
[2] The final inoculum should give a final cell number of 5x10^5 CFU/mL (target range 2x10^5 to 8x10^5 CFU/mL); viable counts should be performed to ensure that the final inoculum is within the specified range (see ISO 20776-1). For all microorganisms, the number of viable cells in the final inoculum depends on the growth phase of the initial culture. This effect is most pronounced for fastidious organisms, which normally needs at least two passages on suitable agar media when retrieved from freezer storage. The dilution required from a McFarland 0.5 suspension depends upon the bacterial species being tested and the method used for inoculum delivery. **A larger volume of the McFarland 0.5 suspension is normally needed for Gram-positive fastidious species and for some Gram-negative fastidious species (e.g. *Actinobacillus pleuropneumoniae and Kingella kingae*).**
[3] 24 h incubation for glycopeptides
[4] For reading instructions, see EUCAST Reading Guide for broth microdilution. For antimicrobial agents with specific reading instructions, see **Table 3**.

Table 2. Antimicrobial agents with supplementation or modification of the broth

Antimicrobial agent	Supplementation/modification
Oxacillin	Medium supplemented to a final concentration of 20 g/L NaCl when tested for *Staphylococcus* spp.
Cefiderocol	Iron-depleted CAMHB, see EUCAST Guidance Document
Lipoglycopeptides (dalbavancin, televancin and oritavancin)	Medium supplemented with 0.002% polysorbate-80
Tigecycline	Medium prepared fresh on the day of use
Daptomycin	Medium supplemented with 50 mg/L Ca^{2+}

Table 3. Antimicrobial agent and species combinations with specific reading instructions (see also EUCAST Reading Guide for broth microdilution)

Antimicrobial agent	Reading MIC endpoints
Cefiderocol	The MIC is read as the first well in which the reduction of growth corresponds to a button of <1 mm or is replaced by the presence of light haze/faint turbidity (≥80% reduction in haze as compared to the growth control). The positive control should show strong growth in the form of a button of >2 mm or heavy turbidity. See EUCAST Guidance Document
Bacteriostatic antimicrobial agents[1]	Disregard pinpoint growth (tiny buttons) when trailing growth occurs
Trimethoprim, trimethoprim-sulfamethoxazole and sulfonamides	Read the MIC at the lowest concentration that inhibits ≥80 % of growth as compared to the growth control.

[1] *E.g*. macrolides, lincosamides, streptogramins, tetracyclines, oxazolidinones, phenicols and fusidic acid.